5.3. Raft Deployment and Post-Incident Survival

Whether a life-raft deploys successfully depends on both current weather severity and how prepared the crew is:

$$P_{\text{raft}} = \text{clip}\left(B_{\text{raft}} - \delta_{\text{wx}}\, W + \delta_{\text{prep}}\, P_{\text{prep}},\ 0,\ 1\right), \tag{12}$$

with $B_{\text{raft}} = 0.90$, $\delta_{\text{wx}} = 0.60$, $\delta_{\text{prep}} = 0.25$. Once in rafts, crew members remain exposed until rescue arrives. Time-to-Arrival (TTA) is:

$$\text{TTA} = \frac{d_{\text{ship,shore}}}{v_{\text{rescue}}} + \Delta t_{\text{mob}}, \tag{13}$$

where $v_{\text{rescue}} = 80\,\text{kn}$ (helicopter) and $\Delta t_{\text{mob}} = 2$ ticks for mobilisation. Survivor probability per raft-tick decreases linearly with current W, so evacuating earlier — when W is still moderate — directly improves survival. This is the causal mechanism behind H2: better mesh coverage → lower E_{wx} → higher P_{prep} → earlier evacuation → lower W at evacuation time → higher survival.

Behavior of this model was motivated by the need to capture the critical importance of evacuation timing in determining survival outcomes. By linking the probability of successful raft deployment to both current weather conditions and crew preparedness, the model reflects the real-world challenges faced by mariners in deciding when to abandon ship. The linear decrease in survivor probability with increasing hazard severity underscores the urgency of timely evacuation, creating a direct causal pathway from the quality of weather intelligence (via E_{wx}) to survival outcomes, which is central to testing H2. The parameters were chosen to produce a realistic range of evacuation success rates and survival probabilities across different scenarios, allowing for meaningful comparisons between the Proposed System and the baselines.

5.4. Collision Detection

Collisions are detected using Shapely circle intersection at a collision radius of $r = 0.15\,\text{nm}$, consistent with near-miss studies on vessel encounter geometry [7]. Each intersection increments both vessels' damage counters; damage above the stability threshold causes a transition to `SUNK` and increments the fatality counter (KPI 1, Table 2).

6. Experimental Design

This section defines the experimental conditions, scenarios, KPIs, and statistical analysis methods used to test the hypotheses.

6.1. Methods Under Test

The three experimental conditions introduced in Section 2 are formally defined as follows:

- **Baseline A (Classical):** No weather intelligence; no preparedness modifier; deterministic COLREGs navigation only. $P_{\text{prep}} \equiv 0$ throughout.
- **Baseline B (Shore-Only):** Shore forecast received via Eqs. 2–3 when in range; no mesh relay; no confidence fusion; no adaptive trust weighting. $\lambda_{\text{ship}} \equiv 0$, so $\hat{W}_{\text{blend}} \equiv \hat{W}_{\text{shore}}$.
- **Proposed System:** Full pipeline — Eqs. 2–7 for forecast fusion, Eqs. 10–12 for behaviour, and hop-limited SOS relay.

An ablation matrix additionally disables each component in turn: mesh off ($\lambda_{\text{ship}} \equiv 0$), evacuation off ($P_{\text{evac}} \equiv 0$), human factors off ($P_{\text{err}} \equiv 0$).

6.2. Scenarios

Table 1 summarises the four scenarios. All share the same $100 \times 100\,\text{nm}$ world; each is implemented as a factory function that configures the model constructor with no scenario-specific branching inside agent code. Every scenario runs 30 independent seeds.

Table 1: *Scenario summary. σ_{shore} is shore forecast standard deviation. All scenarios use 30 seeds.*

Scenario	Key conditions	Hyp.
1 — Calm Passage	20 vessels, $W<0.15$, $\sigma_{\text{shore}}=0.05$. Pass: <0.5 collisions per 1 k ship-hrs.	Calib.
2 — Storm Corridor	25 vessels (30 % Green crew); $W>0.75$ cell on E–W lane; $\sigma_{\text{shore}}=0.18$.	H1
3 — Blind Shore	25 vessels at 40–70 nm; σ_{shore} doubles to 0.36 at tick 40.	H3
4 — Deep Water Rescue	15 vessels >60 nm offshore; W spikes to 0.90 at tick 30; no vessel within shore radio range.	H2, H3

6.3. Key Performance Indicators

Table 2 defines the six KPIs collected per run. KPIs 1–2 test H1, KPIs 3–5 test H2, and KPI 6 tests H3. All are computed from the per-tick Pandas log flushed to Parquet at run end, ensuring reproducibility.

Table 2: *KPI definitions and hypothesis links. ship-hrs = total agent-hours in `ACTIVE` or `EVAC` state.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatal events ÷ ship-hrs $\times 10^3$	H1
2	`collision_per_1k_hrs`	Collision events ÷ ship-hrs $\times 10^3$	H1
3	`survival_ratio`	Survivors / (survivors + fatalities)	H2
4	`avg_tta_hours`	Mean rescue Time-to-Arrival across dispatches	H2
5	`evac_activation_rate`	Fraction of vessels that triggered early evac.	H2, H3
6	`mean_p_prep`	Time-averaged P_{prep} across all vessels	H3

6.4. Statistical Analysis

Hypothesis decisions use the Mann-Whitney U test — appropriate for the bounded, non-normal KPI distributions produced

by a finite-agent simulation. Effect sizes are reported as Cliff's Δ [15]; both $p < 0.05$ and $\Delta > 0.2$ (small effect) are required for H1 and H3 to be confirmed. Confidence intervals are bootstrapped over the 30 seeds. The same seed sequence is applied to all conditions to eliminate seed-selection bias from pairwise comparisons.

7. Discussion

The target reference rate for calibration — 5 fatal events per 1 000 ship-hours under Baseline A in Scenario 2 (Table 1) — is grounded in AIS-derived near-miss encounter statistics [6]. If H1 or H2 fail statistically, the mitigation is to recalibrate α, β, γ in Eq. 11 or widen σ_{shore} before final reporting; design parameters were chosen conservatively to make this adjustment feasible without invalidating the comparison.

The two-hop relay is sized deliberately: with 15 nm vessel radio range and vessels distributed along the 100 nm east–west tanker lane, three ships can collectively bridge the 45 nm shore shadow tested in Scenario 4 (Table 1). Removing the hop limit (unbounded flooding) would close the gap trivially and obscure the protocol's practical value; the two-hop constraint models a real latency and battery trade-off in maritime IoT deployment.

The simulation is kept compact — up to 50 agents and a 50×50 weather grid — to remain within a 16 GB RAM ceiling and to produce visually legible scenario track maps in a live Streamlit dashboard. The framework is nonetheless extensible: lane geometry, agent counts, weather severity, and σ_{shore} are all constructor parameters, and additional experiment configurations can be added as factory functions without modifying any agent logic.

8. References

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